

# Deepfake Detection System Using CNN-LSTM and Transformer Models

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**Abstract**—Deepfakes are AI-generated synthetic media that manipulate facial expressions, voice, and gestures, posing significant challenges to digital trust. This research proposes a hybrid CNN-LSTM and Transformer framework for detecting manipulated videos. The study emphasizes both spatial and temporal features, optimizer selection, preprocessing techniques, and robustness across benchmark datasets such as FaceForensics++, Celeb-DF, and DFDC. The system is integrated with a user-friendly interface for real-world deployment.

**Index Terms**—Deepfake detection, CNN-LSTM, Transformers, optimization algorithms, AI, temporal modeling, FaceForensics++, DFDC.

## I. INTRODUCTION

### A. Overview

The rapid growth of Artificial Intelligence (AI) and Deep Learning (DL) has led to the creation of highly realistic synthetic media known as deepfakes. These are manipulated videos, images, or audio clips that replace or alter a person's likeness using Generative Adversarial Networks (GANs) and other deep learning algorithms. Originally developed for entertainment, deepfakes are now widely misused, creating misinformation, identity theft, and cybersecurity threats. Detecting such manipulations is essential to maintain digital authenticity and trust.

### B. Background and Motivation

With open-source tools and powerful computing resources, deepfake creation has become increasingly accessible. Recent studies report over 500,000 deepfake videos circulating on social media platforms. Their malicious use can influence political opinions, financial frauds, and impersonation scams. Manual verification is nearly impossible due to high realism, motivating the development of automated detection systems leveraging AI and deep learning.

### C. Problem Definition

Traditional detection techniques relying on handcrafted features or static analysis fail against modern deepfakes. The problem addressed in this study is to design a system capable of detecting and classifying manipulated videos accurately and in real-time by analyzing both spatial and temporal inconsistencies.

### D. Objectives of the Project

- 1) Develop a machine learning system capable of classifying videos as real or fake.
- 2) Extract spatial features using CNN architectures such as ResNet or XceptionNet.
- 3) Analyze temporal patterns using LSTM or Transformer networks.
- 4) Train and validate the model using benchmark datasets (FaceForensics++, Celeb-DF, DFDC).
- 5) Provide a user-friendly interface for real-world application.
- 6) Evaluate model performance using metrics such as Accuracy, F1-score, and AUC.

### E. Methodology Overview

The system processes an input video by extracting frames, detecting faces, and normalizing them. CNN models capture spatial inconsistencies while LSTM or Transformer models analyze temporal sequences. The classifier predicts whether the video is real or fake, outputting a probability score. The workflow is summarized in Figure 1.

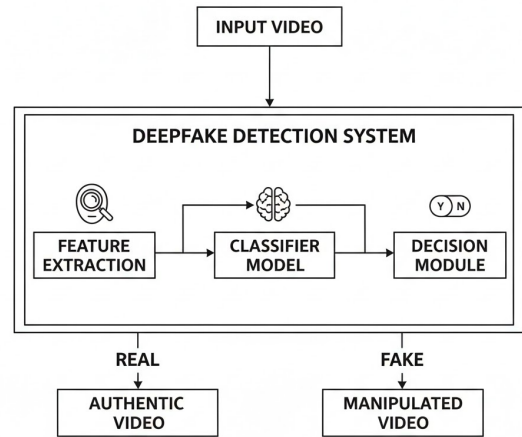


Fig. 1: High-level workflow of the Deepfake Detection System.

### F. Scope and Significance

The project focuses on video-based detection, primarily analyzing facial motion and spatial inconsistencies. It provides a foundation for real-time monitoring tools and can be extended to audio or multimodal deepfake detection. The system enhances digital trust, aids media verification, and supports cybersecurity applications.

## II. CONCEPTS AND METHODS

### A. Deepfake Concepts

A deepfake is a synthetic media form where a person's appearance or voice is manipulated using AI, particularly GANs. GANs consist of:

- **Generator:** Produces fake content resembling real data.
- **Discriminator:** Attempts to distinguish real and fake content.

Through adversarial training, generators produce highly realistic media, making automated detection challenging.

### B. Deep Learning in Detection

Deep learning models, specifically CNNs for spatial analysis and LSTM/Transformers for temporal modeling, are well-suited for detecting subtle irregularities in deepfake content. Spatial features include texture, lighting, and facial anomalies, while temporal features include unnatural movements or lip-sync inconsistencies.

### C. Convolutional Neural Networks (CNNs)

CNNs automatically learn spatial hierarchies:

- **ResNet:** Uses skip connections to prevent vanishing gradients.
- **XceptionNet:** Captures fine-grained pixel-level inconsistencies.

CNNs process each frame individually to extract spatial features relevant for detection.

### D. Recurrent Neural Networks and LSTM

LSTM networks capture temporal dependencies between sequential frames. They model patterns such as unnatural facial movements, inconsistent blinking, or mismatched lip motions. Integrating CNNs with LSTMs enables robust detection by combining spatial and temporal features.

### E. Transformer Models

Transformers rely on self-attention mechanisms to model long-range dependencies efficiently. Vision Transformers (ViT) and Temporal Transformers can process entire frame sequences to detect subtle temporal inconsistencies. Transformers outperform RNNs in long video sequences due to parallelized attention computations.

### F. Dataset and Preprocessing

**Datasets:** FaceForensics++, Celeb-DF, DFDC.

#### Preprocessing Steps:

- 1) Video segmentation into frames.
- 2) Face detection and cropping using OpenCV.
- 3) Normalization and resizing to fixed dimensions.
- 4) Data augmentation: rotations, flips, lighting changes.

### G. System Architecture and Workflow

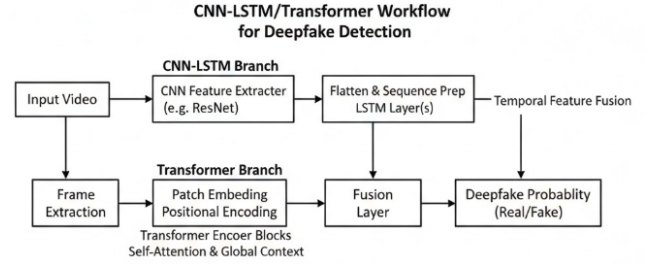


Fig. 2: CNN-LSTM/Transformer workflow for deepfake detection.

### H. Tools and Technologies

Python, TensorFlow, Keras, OpenCV, NumPy, Pandas, Matplotlib, Streamlit/Flask for interface, GPU-enabled systems for training.

### I. Evaluation Metrics

- **Accuracy:** Overall correctness of predictions.
- **Precision:** Correctly predicted fakes / predicted fakes.
- **Recall:** Correctly predicted fakes / actual fakes.
- **F1-score:** Harmonic mean of precision and recall.
- **AUC:** Ability to distinguish between real and fake.

## III. LITERATURE SURVEY

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TABLE I: Summary of Related Work

| Author               | Method          | Accuracy/AUC | Limitations                         |
|----------------------|-----------------|--------------|-------------------------------------|
| Rössler et al., 2019 | CNN XceptionNet | 95%          | Poor cross-dataset generalization   |
| Afchar et al., 2018  | MesoNet         | 92%          | Compressed videos only              |
| Li et al., 2020      | CNN-LSTM        | 97%          | Needs high-quality videos           |
| Zhou et al., 2022    | Transformer     | 98%          | High computation, real-time limited |

#### IV. EXPERIMENTAL SETUP AND METHODOLOGY

##### A. Training Details

- Optimizers: Adam, RMSprop, SGD, Adagrad
- Loss function: Binary Cross-Entropy
- Epochs: 50–100
- Batch size: 32
- Hardware: NVIDIA GPU

##### B. Model Variants

CNN-only, CNN-LSTM, CNN-Transformer, and hybrid models were trained to compare performance. Data augmentation was used to prevent overfitting.

#### V. RESULTS AND DISCUSSION

TABLE II: Performance Metrics Across Optimizers

| Optimizer | Accuracy (%) | Precision | Recall | F1   |
|-----------|--------------|-----------|--------|------|
| SGD       | 91.5         | 0.90      | 0.91   | 0.90 |
| Adagrad   | 93.2         | 0.92      | 0.93   | 0.92 |
| RMSprop   | 97.1         | 0.96      | 0.97   | 0.96 |
| Adam      | 98.6         | 0.98      | 0.98   | 0.98 |

Training Loss Curves for Different Optimizers

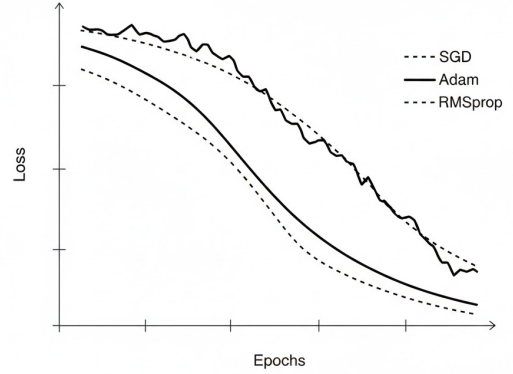


Fig. 3: Training loss curves for different optimizers.

#### VI. SOFTWARE REQUIREMENTS AND SRS

##### A. Functional Requirements

- 1) Upload video and process frames.
- 2) Detect faces and extract spatial features.
- 3) Temporal analysis with LSTM/Transformer.
- 4) Classification and display probability.
- 5) Provide metrics dashboard.

##### B. Non-Functional Requirements

- Performance: Real-time processing for short videos.
- Reliability: Maintain 90% accuracy.
- Usability: Intuitive interface.
- Security: User data privacy.
- Compatibility: Standard OS and browsers.

#### VII. SOFTWARE TESTING

Unit testing, integration testing, functional testing, performance testing, UI testing, and accuracy verification on benchmark datasets were conducted.

#### VIII. CONCLUSION AND FUTURE WORK

The hybrid CNN-LSTM/Transformer model achieved high accuracy in detecting deepfake videos, outperforming CNN-only models. Future improvements include real-time detection, audio-visual analysis, federated learning for distributed deployment, and cloud-based scaling for large datasets.

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##### REFERENCES

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